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Use of generative artificial intelligence models in healthcare – a scoping review

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Abstract

Background This scoping review aims to identify and assess the use of pre-trained, fine-tuned, and multimodal Generative Artificial Intelligence (GenAI) models in healthcare, focusing on their application areas, reported benchmarks, and environmental impact.

Methods A comprehensive search was conducted in PubMed, Scopus, and Cochrane Central Register of Controlled Trials databases, supplemented by manual searches. Studies were selected based on inclusion criteria that focused on the application of GenAI models in healthcare. The review followed PRISMA-SCR guidelines and synthesized data on the type of models, their tasks, accuracy, and carbon footprint.

Results Out of 7,351 initial studies, 24 studies met the inclusion criteria. The models were utilized in tasks such as classification, medical Visual Question Answering (VQA), and conversational tasks, with reported accuracies ranging from 70.6% to 99.9%. Fine-tuned models showed superior accuracy with shorter training times compared to pre-trained models. Only one study reported the carbon footprint, revealing a significant gap in environmental reporting. Multimodal AI models, although less common, demonstrated promising results in handling complex healthcare data.

Conclusion GenAI models hold significant potential in healthcare. However, the review highlights the underutilization of multimodal models and the lack of reporting on carbon footprints. Future researches should focus on optimizing task-specific GenAI applications while addressing their environmental impact.

Keywords Generative Artificial Intelligence, Large Language Models, Healthcare, Machine Learning, Deep Learning

Background

Artificial Intelligence (AI) has become a cornerstone of modern technology, driving innovation and transforming the way we approach problem-solving [1]. Over the years, growth of AI has expanded from simple Machine Learning (ML) and Deep Learning (DL) models to complex large-scale models designed to improve human efficiency [2]. A prime example of this shift is the emergence of Generative Artificial Intelligence (GenAI) models such as OpenAI's *ChatGPT*, *Gemini*, and *MedPaLM*, although this list is far from exhaustive.

GenAI can be broadly divided into Large Language Models (LLMs) and multimodal models, both of which can be further categorized into pre-trained and fine-tuned models. A pre-trained model is trained on large datasets from various fields, allowing it to learn underlying patterns and context across different domains [3]. However, to make these models truly effective in specialized areas such as healthcare, they undergo a fine-tuning

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process. During fine-tuning, these pre-trained models are further refined on specific, domain-related data [4]. This allows sophisticated applications in healthcare by answering complex questions, performing data analysis and even assisting in decision-making processes [5].

Recently, *Med-PALM2*, a medical GenAI, which was fine-tuned with various medical related Question Answer (QA) datasets emerged as a key-player in answering US Medical Licensing Examination (USMLE) style questions [6]. Similarly, *Nova-2* model was released which is a speech-to-text model specifically designed to transcribe medical terms accurately into text [7]. In addition to fine-tuning, another approach is the use of multimodal models which are capable of processing information from diverse data modalities [8]. Multimodal models are particularly valuable in healthcare because they can interpret both visual and textual data, assisting in tasks like diagnosing diseases from medical scans while considering patient records [9].

Despite their promise, the training and deployment of GenAI models are highly resource-intensive consuming substantial amounts of energy [10]. These demands are driven by three main factors: lengthy model training involving billions of parameters, inference generation for each user prompt, and data storage on cloud servers. GenAI models also rely heavily on computational infrastructure, primarily composed of Graphics Processing Units (GPUs) to process large amounts of data and create content. While GenAI models have brought great advancements, their use adds to environmental concerns, increasing both carbon emissions and water consumption [11]. For instance, training GPT-3 with 175 billion parameters consumed 1,287 megawatt hours and emitted 552 tons of $CO2$, equivalent to the emissions of 100 cars over their entire lifetime [12]. Similarly, training GPT-3 in data centers can directly evaporate 700,000 litres of clean, fresh water required for cooling systems [13].

Energy requirements for training GenAI models can be significantly reduced by focusing on fine-tuning pre-existing models instead of training new models from scratch [14]. This approach leverages foundational knowledge already learned during pre-training, conserves energy, and aligns with sustainability goals by mitigating the carbon and water footprints of AI development [15]. Likewise multimodal models are advantageous in this regard because they consolidate different data types within a single model, reducing the need to train multiple specialized models [8].

GenAI holds tremendous potential to transform healthcare by analyzing large volumes of medical data, aiding in clinical decisions, and enhancing healthcare delivery. This scoping review aims to identify the availability and use of GenAI models in healthcare while evaluating the different types of models, including pre-trained, fine-tuned, and multimodal models, to gain a deeper understanding of their applications. It also seeks to determine the most commonly reported benchmarking datasets and to assess how many studies have addressed environmental sustainability by reporting the carbon footprints associated with their respective models. To the best of the authors' knowledge, no review to date has comprehensively combined these aspects within a single framework.

Materials and methods

This scoping review was conducted using a pre-determined protocol following the PRISMA extension for scoping review guidelines (PRISMA-SCR). The protocol for this review can be accessed through the Open Science Framework (OSF) platform (https://osf.io/hkagz).

Database search

Search strategy

A 3-pronged approach was applied using a predefined search strategy with a focus on GenAI models such as pre-trained, fine-tuned and multimodal models. In collaboration with a medical information specialist (Librarian, The Aga Khan University Hospital) the search strategy was finalized on 20th August 2024. Initially the authors conducted a pilot search on the basis of relevant keywords which were used to formulate a final search strategy. Electronic databases such as PubMed, Scopus, and the Cochrane Central Register of Controlled Trials (CENTRAL) were used to retrieve relevant studies addressing the research question. The complete search strategy is presented in Table 1.

Eligibility criteria

Inclusion criteria

1) Original peer-reviewed studies and preprints focusing on pre-trained, fine-tuned and multimodal models utilized in healthcare.
2) Studies published in the English language.
3) Studies published in the last 10 years.

Exclusion criteria

1) Review articles, editorials, letter to the editors, conference proceedings and commentaries.
2) Studies published in languages other than English.

Screening process

The initial list of citations (7351) was imported into EndNote version 21.0 (Clarivate Analytics), and duplicate records were removed. The remaining citations (7042) were then equally divided between two reviewers (3521

Table 1 Search strategy for studies retrieval

Database	Search Terms	Count
PubMed	("Artificial Intelligence" OR "Large Language Models" OR "LLM" OR "Transformer Models" OR "Generative AI" OR "GPT" OR "Chatbot*" OR "Generative Pre-Trained Transformer" OR "Natural Language Processing" OR "Retrieval Augmented Generation" OR "Pretrained" OR "Medical AI" OR "Generative Pre-Trained Transformer" OR "AI Chatbot" OR "Fine Tuning" OR "Pre-trained Model" OR "Prompt Engineering") AND ("Healthcare"[Mesh] OR "Medical"[Mesh] OR "Clinical"[Mesh] OR "Hospital"[Mesh] OR "Patient care"[Mesh] OR "Medicine"[Mesh] OR "Medical/Health Services" OR "Healthcare Systems") AND ("Application" OR "Impact" OR "Evaluation" OR "Ethical concerns" OR "Decision-making" OR "Clinical outcomes" OR "Diagnosis" OR "Treatment" OR "Patient interaction" OR "Data Augmentation" OR "Specificity" OR "Sensitivity" OR "Accuracy")	357
Scopus	("Artificial Intelligence" OR "Large Language Models" OR "LLM" OR "Transformer Models" OR "Generative AI" OR "GPT" OR "Chatbot*" OR "Generative Pre-Trained Transformer" OR "Natural Language Processing" OR "Retrieval Augmented Generation" OR "Pretrained" OR "Medical AI" OR "Generative Pre-Trained Transformer" OR "AI Chatbot" OR "Fine Tuning" OR "Pre-trained Model" OR "Prompt Engineering") AND ("Healthcare" OR "Medical" OR "Clinical" OR "Hospital" OR "Patient care" OR "Medicine" OR "Health Services" OR "Healthcare Systems") AND ("Decision-making" OR "Clinical outcomes" OR "Patient interaction" OR "Data Augmentation" OR "Performance Metrics")	4401
CENTRAL	("Artificial Intelligence" OR "Large Language Models" OR "LLM" OR "Transformer Models" OR "Generative AI" OR "GPT" OR "Chatbot*" OR "Generative Pre-Trained Transformer" OR "Natural Language Processing" OR "Retrieval Augmented Generation" OR "Pretrained" OR "Medical AI" OR "Generative Pre-Trained Transformer" OR "AI Chatbot" OR "Fine Tuning" OR "Pre-trained Model" OR "Prompt Engineering") AND ("Healthcare" OR "Medical" OR "Clinical" OR "Hospital" OR "Patient care" OR "Medicine" OR "Health Services" OR "Healthcare Systems") AND ("Decision-making" OR "Clinical outcomes" OR "Patient interaction" OR "Data Augmentation" OR "Performance Metrics")	2593
Manual	("Generative Artificial Intelligence" OR "GenAI" OR "Large Language Models" OR "LLM" OR "Transformer Models" OR "GPT" OR "ChatGPT" OR "Gemini" OR "PaLM" OR "Medical AI" OR "Pre-trained model" OR "Fine-tuned model" OR "Multimodal model") AND ("Healthcare" OR "Medicine" OR "Clinical" OR "Patient care" OR "Diagnosis" OR "Treatment" OR "Decision-making")	13

each), who independently screened them. Titles were initially reviewed based on the predefined inclusion and exclusion criteria, followed by abstract screening when the title alone was insufficient to determine eligibility. After excluding irrelevant studies (7031), 11 potentially relevant studies were identified.

Manual search

Manual search was conducted by (S.M.F.A) through Google Scholar and reference snowballing from the full-text articles retrieved. The search string used for manual search is presented in Table 1. All articles identified through the manual search were ored in a separate folder for record-keeping. The search was performed up to the same end date as the main database search, i.e. 31st August 2024. The manually identified articles were subsequently reviewed and discussed by (A.N and A.L) to determine their relevance. Any duplicates found between the manual and database searches were removed from the manual search folder, and only unique articles were retained in the EndNote library.

Full texts of the 11 studies identified through the database search and 13 from the manual search were retrieved. Data from the final set of included studies were extracted using the data extraction proforma described in the subsequent section.

Data synthesis

For data synthesis, a predesigned data extraction form was drafted using Google Sheets (Google©), with the following variables: Author, year of publication, study title, study link, study setting, study design, study theme, objective of the study, dataset source, data type, sample size, model name, base model name, tasks of the model, technique (pre-training and/or fine-tuning), multimodal model, number of GPUs, carbon emissions during model training, performance metrics and benchmarking datasets.

Information based on these variables was extracted from the papers that met the eligibility criteria. The extraction was performed independently by two authors (A.N and A.L), who equally divided the final set of articles between them. Entries completed by one author were cross-checked by the other. Any disagreements were resolved by a third author (S.M.F.A). Considering the heterogeneity and variability in the models' tasks, a narrative synthesis approach was applied for data synthesis. The results are organized according to the type of model: pre-trained, fine-tuned, and multimodal. Each category was further classified based on the specific tasks performed by the models, such as classification, conversational tasks, and semantic similarity. Key concepts and terminologies used to describe the results are defined in Additional file 1 (Operational Definitions).

Results

Search results and study inclusion

The initial search yielded a total of 7,351 studies up until August 31st, 2024. After removing 309 duplicate records, 7,042 studies remained for further screening. A thorough selection process then excluded 7,031 articles for various reasons, including irrelevant titles, focus on general-purpose GenAI models, or being models other than GenAIs, such as CNNs. This resulted in 11 articles for inclusion from the databases. Additionally, 13 more studies were identified and included through manual search. This search was conducted using Google Scholar and through the snowballing of references from previously published articles. Consequently, a total of 24 studies were selected for the scoping review, as illustrated in Fig. 1 (PRISMA).

Study characteristics

The characteristics of the included studies are summarized chronologically in Table 2. This scoping review encompasses studies published between 2019 and 2024. Most of the studies were conducted in China ($n = 6$) [16–21] and United States of America [22–27]. However, some were multi-center studies ($n = 6$) [6, 28–31].

Variable data sources were used across the studies. A total of 6 studies [6, 21, 27, 30, 32, 33] used data from Visual Question Answering (VQA) datasets, while 5 studies [20, 26, 34–36] used PubMed as their data source. Additionally, 5 studies [17, 18, 22, 24, 28] used data from Electronic Health Records (EHRs). The remaining 8 studies had variable data sources such as Vaccine Ontology (VO), Unified Medical Language System (UMLS) and Diabetic Retinopathy Detection (DRD) [16, 19, 23, 25, 29, 31, 37, 38]. The sample sizes in the included studies varied significantly based on the type of data used. Studies focusing on textual data used samples ranging from 2,054 sentence pairs for clinical text analysis [22] to over 36,950 abbreviations [29]. Studies using images often had larger data samples, ranging from 149,000 images [21] to 15 million images [31].

Pre-trained and fine-tuned models

A total of 13 studies pre-trained and then fine-tuned their models on large scale clinical datasets for various classification and conversational tasks [17–19, 21, 24, 26, 27, 30, 31, 33–35, 37]. Performance metrics of the models evaluated in the included studies are summarized chronologically in Table 3.

Classification tasks

Out of the 13 studies reviewed, 5 focused [17–19, 24, 37] on classification tasks, including specialty predictions and categorization of disease severity [17, 19]. A total of 4 out of 5 studies reported the accuracies for these classification tasks which ranged from 86% to 93% with *Med-PLM* model exhibiting the highest accuracy [19]. The study which did not report the accuracy provided the F-1 score of 0.99 [17]. A total of 2 studies employed *BERT* as

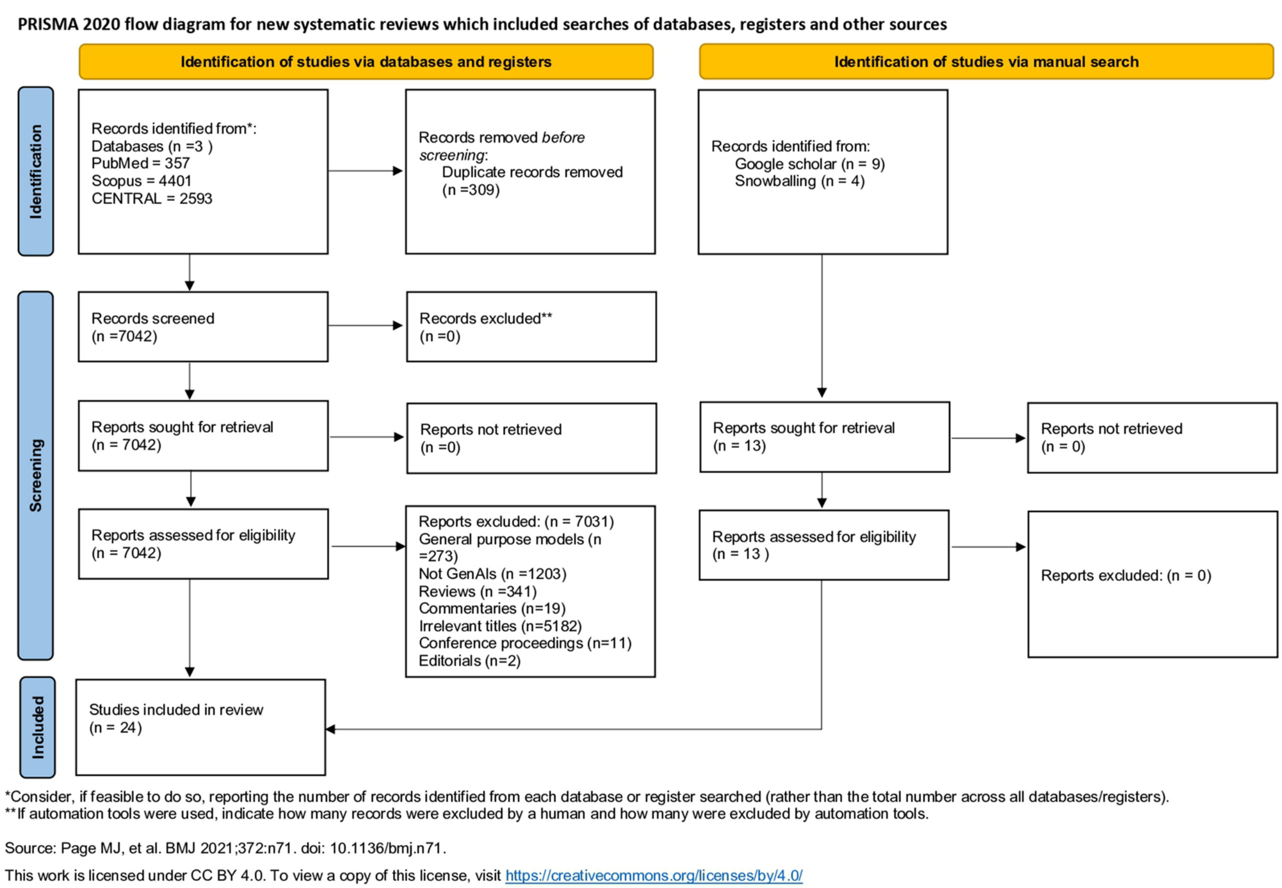

Fig. 1 PRISMA flow chart

Table 2 Summary characteristics of the included studies ($n = 24$)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Study theme**	**Dataset source**	**Data type**	**Sample size**	**Model name**	**Base model name**	**Task of model**
Task	Yoojong Kim et al. 2022	Development of a model to predict medical speciality	NAVER portal + web crawling	medical consultation question texts for 27 medical specialty labels	50,454 questions	Korean Medical BERT (KM-BERT)	NM	Medical specialty prediction
	Jia Li et al. 2022	BERT-based approach for classifying actionable radiology reports of tinnitus patients	EHR) data from patients who were admitted with tinnitus and received temporal bone CT	Temporal bone (CT) reports	Total of 5864 temporal bone CT reports	BERT FINETUNE, BERT IDPT	BERT	BERT-based framework for tinnitus diagnosis from Chinese radiology reports
	Jianfu Li et al. 2023	Developed a cascaded framework using clinical trials, UMLS, and VO.	VO, UMLS, AACT	Vaccine termcons (text data)	550 vaccine terms	CTPubMedBERT + SAPBERT + VO model	PubMedBERT	Enhance performance of domain-specific language models for automated mapping of VO.
	Sicen Liu 2023	Multimodal learning that combines structured and unstructured EHR data	MIMIC-III dataset	Vital signs, medications, procedure codes, diagnostic codes, and clinical narratives	35,164 patients records	MedM-PLM	NM	To learn enhanced EHR representations and explore the interaction of two modalities
	Zhangzhong Gu et al. 2024	LLM-based CNER model, and automated NIHSS scoring.	CCKS2019, CSCR from EHRs	Admission and discharge notes, Admission and discharge records	Admission and discharge notes = 1133, Admission records = 1067, Discharge records = 864	Domain-adaptive pre-trained LLM (Cli-Roberta) and the CNER model	Roberta-wwm	To extract stroke related entities directly from Chinese EHRs

Table 2 (continued)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Study theme**	**Dataset source**	**Data type**	**Sample size**	**Model name**	**Base model name**	**Task of model**
Conversational	Jinhyuk Lee et al. 2019	First domain-specific BERT based model pre trained on biomedical corpora	PubMed & PMC	Biomedical Corpora	4.5 B Words PubMed & 13.5B words PMC	BioBERT	BERT(zero weights)	General Purpose pretrained medical LLM
	Michihiro Yasunaga et al. 2022	MLM and new proposal, document relation prediction.	PubMed abstracts with citation links	PubMed abstracts	21 GB	BioLinkBert (Large) / (110Mparams) and (340Mparams)	NM	General Purpose pretrained medical LLM
	Pengfie Li et al. 2023	Self supervised method for pre-training on medical image caption dataset, and fine-tuned to downstream medical VQA tasks	Pre-training: ImageCLEF2022, Fine-tuning: VQA-RAD, Slake, Path, VQA	Closed and open ended questions	NM	M2I2	NM	solve medical VQA tasks as generative processes.
	Zeming Chen et al. 2023	Development of a pre-trained LLM specifically for healthcare	PubMed articles, abstracts, and internationally-recognized medical guidelines	PubMed abstracts	NM	MEDITRON-70B	NM	General purpose pre-trained LLM
	Gang Liu et al. 2024	Parameter efficient framework for fine-tuning MLLMs	ROCO, CLEF2022, MEDICAT, and MIMIC-CXR	Various datasets	NM	PeFoMed	NM	Med-VQA and MRG
	Sheng Zhang et al. 2024	To facilitate future research in multimodal biomedical AI.	PMC-15 M	Biomedical image-text pairs,	15 million image-text pairs	BiomedCLIP	NM	Image classification & VQA
	Kai Zhang et al. 2024	Development of BiomedGPT, and its evaluation in biomedical tasks.	MIMIC-III, MIMIC-CXR, VQA-RAD, BLUE, SLAKE 1.0	Text and Images	592,567 images, 183 million text sentences, 46,408 object-label pairs, and 271,804 image-text pairs	BioMedGPT	NM	Radiology VQA, report generation, and summarization
	Xiaoman Zhang et al. 2024	Generative model for MedVQA by aligning a vision encoder with a LLM	PMC-VQA	Images	149k images	MedVInT, MedVInT-TE, MedVInT-TD	Pre-trained vision encoder with a LLM through visual instruction tuning	Perform generative-based MedVQA.
Fine-tuned								

Table 2 (continued)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Study theme**	**Dataset source**	**Data type**	**Sample size**	**Model name**	**Base model name**	**Task of model**
Classification	Brian J Ferrell et al. 2022	Application of attention-based models and transformer-based models	IRB database at VCU, hand-classified protocols from 2013–2019	Research protocols from IRB submissions	Initial sample of 280 manually classified protocols, with over 6000 unlabeled protocols used for model application	Bi-LSTM with attention layer, BERT, Bio + Clinical BERT, XLM-RoBERTa	GloVe (for embeddings), BERT-based models	To classify research protocols based on their level of community engagement
	Aron Henriksson et al. 2023	Multimodal learning that combines structured and unstructured EHR data.	EHR data	Structured and Unstructured EHR data	20,000 EHR	Clinical KB-BERT	BERT	Predict COVID-19 patient outcomes at day 30th, discharge, and readmission.
	Ethan Schonfeld et al. 2024	Machine learning applications CV, NLP, GWAS for predicting surgical risks and outcomes in neurosurgery	Stanford Research Repository (clinical data), VinDr-SpineXR dataset (pre-trained model),	Radiographs (imaging), clinical notes (text), genomic data (SNPs)	209 patients (1549 radiographs), 540 surgical ASD patients (GWAS)	CNN, Gatortron (LLM), GWAS using PLINK	Gatortron (for LLM)	Predict post-operative complications
	Manda Hosseini et al. 2024	Utilized LLMs and a GM for disambiguation in clinical contexts.	CASI dataset	Abbreviations from CASI dataset	36,950 instances	BlueBERT + Transformer + BIOGPT	Blue-BERT, BIOGPT	To generate clinical text that represent the meanings of abbreviations

Table 2 (continued)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Study theme**	**Dataset source**	**Data type**	**Sample size**	**Model name**	**Base model name**	**Task of model**
Conversational	Joao Luis Zeni Montenegro et al. 2022	To promote pregnant women's health literacy through a conversational agent called HoPe	Health documents related to pregnancy care.	paragraphs/sentences	1.500 paragraphs/sentences	HoPE	BERTimbau model	Provide information to patients related to pregnancy.
	Weihao Gao et al. 2023	Use of a multimodal language-and-vision assistant for medical image classification, segmentation, and diagnosis of ophthalmic diseases	Fundus images from public datasets like DRD, DDR, REFUGE, ORIGA650, BAIDU PALM, and private datasets.	Multimodal data combining medical images (fundus images) and medical dialogues	Over 20k images	OphGLM	ChatGLM (6.2 billion parameters)	Disease diagnosis and lesion segmentation from fundus images, along with medical Q&A dialogues
	Chunyuan Li et al. 2023	Multimodal conversational assistant for biomedical image understanding	PMC-15 M (PubMed Central dataset with 15 million biomedical image-text pairs)	Biomedical image-text pairs	600,000 image-text pairs for Stage 1; 60,000 pairs for Stage 2	LLaVA-Med	LLaVA (Large Language-and-Vision Assistant)	Vision-language conversational assistant for biomedical images
	Karan Singhal et al. 2023	Improving the way LLM answers as compared to clinicians' answers	MedQA, MedMCQA HealthSearchQA, LiveQA and MedicationQA.	MCQs and Long form questions	NM	Med-PaLM 2	PaLM 2	General healthcare model
	Yizhen Luo et al. 2023	BioMedGPT allows users to communicate with diverse biological modalities through free text	Full text only biomedical papers from PubMed Central (PMC)-ID and PubMed ID	Full text papers	4.2 million full text papers	BioMedGPT-10B	Meta AI - Llama2	Healthcare LLM with protien and Molecule Encoder
	Hyunjae Kim et al. 2024	Development of a medical SOTA model in SLM	Medical Books, MedMCQA, Clinical Notes, Free form QA etc.	Chain of Thoughts, instructions based	460,625 training examples	Meerkat-7B	Mistral-7B-v0.1	General purpose healthcare LLM
Semantic similarity	David Chang et al. 2021	Incorporating domain knowledge for clinical semantic textual similarity assessment	MedSTS dataset+ EHRs	Text pairs	2054 sentence pairs	ClinicalBERT-DA + GCN_snomedkge	ClinicalBERT	Assessing clinical semantic textual similarity

NM: Νοτ μεντιονεδ, NLP: Natural Language Processing, GCN: Graph Convolutional Networks, CT: Computed Tomography HoPE: Healthcare Obstetric in Pregnancy, EHR: Electronic Health Records, MIMIC-III: Medical Information Mart for Intensive Care III), VO: Vaccine Ontology, MIM: Masked Image Modeling, MLM: Masked Language Modeling, CV: Computer Vision, CSCR: Chinese Stroke Clinical Records, LLMs: Large Language Models GM: Generative Model, MedVQA: Medical Visual Question Answering, MRG: Medical Report Generation, VQA: Visual Question-Answering, MedVInT: Medical Visual Instruction Tuning, DRD: Diabetic Retinopathy Detection

Table 3 Diagnostic performance of different AI-models in the included studies ($n = 24$)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Technique (Pre-trained & fine-tuned)**	**Multi-modal (Yes/No)**	**GPU used for training**	**Training hours**	**Accuracy**	**Precision**	**Benchmark**
	Yoojong Kim et al. 2022	Pre-trained and fine-tuned	No	NM	NM	0.891	0.551	GLUE, SQuAD, SWAG, NER
	Jia Li et al. 2022	Pre-trained & fine-tuned	No	1 x Nvidia GTX1070	32 min / Epoch, total training epoch 10 in total 320 min for IDPT	FINETUNE = 0.868, IDPT = 0.948	0.843	DL base models, BERT-base model and BERT variants
	Jianfu Li et al. 2023	Pre-trained and fine-tuned	No	8 x Nvidia A100	NM	90.0	0.90	Baseline SAPBERT model
	Sicen Liu 2023	Pre-trained and fine-tuned	No	Two GeForce RTX 3090 GPUs	NM	Medication recommendation: 92.93 30 day readmission: 68.77 ICD coding: 35.22	NM	MIMIC-III
	Zhangzhong Gu et al. 2024	Pre-trained	No	NM	NM	NM	NM	CCKS2019
Conversational	Jinhyuk Lee et al. 2019	Pre-trained and fine-tuned	No	8 x V100 GPUs	552	NM	88.2	NCBI Disease, BC2GM, 2010 i2b2,VA, BC5CDR-disease, BC4CHEMD, BC5CDR-chemical, JNLPBA, LINNAEUS, Species-800
	Yasunaga et al. 2022	Pre-trained	No	8 x A100 80G	340 Mparams = 504 110Mparams = 168	BLURB:84.30, MedQA-USMLE: 44.6, MMLU-Professional medicine: 50.7 BLURB: 83.39, MedQA-USMLE: 40.4	NM	BLURB, MedQA-USMLE, MMLU-Professional medicine BLURB, MedQA-USMLE
	Pengfie Li et al. 2023	Pre-trained and fine-tuned	Yes	NM	NM	VQA-RAD = 76.8 Slake = 62.2, PathVQA = 81.2	NM	VQA-RAD, Slake, PathVQA
	Zeming Chen et al. 2023	Pre-trained and fine-tuned	No	128 x A100 80G	332	MMLU-Medical: 71.5, PubMedQA: 79.8, MedMCQA: 53.3, MedQA: 52.0, MedQA-4-Option: 59.8	NM	MMLU-Medical, PubMedQA, MedMCQA, MedQA, and MedQA-4-Option
	Gang Liu et al. 2024	Pre-trained and fine-tuned	Yes	TeslaA40GPUs	NM	85.30	NM	VQA-RAD, Slake, PathVQA and IU-Xray
	Sheng Zhang et al. 2024	Pre-trained and fine-tuned	Yes	16NVIDIAA100GPUs or 16NVIDIAV100GPUs	NM	86.1	NM	MAML, CLIP and PubMedCLIP
	Kai Zhang et al. 2024	Pre training and fine-tuned	Yes	10 NVIDIA A5000 GPUs	base, medium, and small-scale models take approximately 87, 32, and 9 h, respectively	BioMedGPT-B = 89.5, BioMedGPT-S = 77.8, BioMedGPT-M = 89.2	NM	VQA-RAD, SLAKE, and Path-VQA, AIBL, MIRIAD, LIDC
	Xiaoman Zhang et al. 2024	Pre-trained	Yes	8 NVIDIA A100 GPUs	NM	80	NM	VQA-RAD, SLAKE, ImageClef-VQA-2019,
Fine-tuned								

Table 3 (continued)

	Pre-trained and Fine-tuned							
	Author and year of publication	**Technique (Pre-trained & fine-tuned)**	**Multi-modal (Yes/No)**	**GPU used for training**	**Training hours**	**Accuracy**	**Precision**	**Benchmark**
Classification	Brian J Ferrell et al. 2022	Fine-tuned	No	NM	Bi-LSTM model was trained for 12 h	0.995	NM	BERT, Bio + Clinical BERT, XLM-RoBERTa
	Aron Henriksson et al. 2023	Fine-tuned	No	NVIDIA Tesla T4 GPU	NM	NM	0.86	Unimodal models (structured data only vs. unstructured data only), late fusion models
	Ethan Schonfeld et al. 2024	Pre-trained and fine-tuned	Yes	NM	NM	NM	0.46	Predict post-operative complications
	Manda Hosseini et al. 2024	Fine-tuned	No	NM	NM	99.53	0.99	CASI
Conversational	Joao Luis Zeni Montenegro et al. 2022	Fine-tuned	No	Tesla P100	NM	NM	0.906	ASSIN2, STS
	Weihao Gao et al. 2023	Fine-tuned	Yes	NM	NM	DR classificatio: 97 Glaucoma: 94, PALM: 99.8, AMD: 98.4, Tumor: 99.9	NM	Visual Med-Alpaca, LLaVA-Med
	Chunyuan Li et al. 2023	Fine-tuned	Yes	8 A100 GPUs	Less than 15 h	91.21	NM	VQA-RAD, SLAKE, and PathVQA
	Karan Singhal et al. 2023	Fine-tuned	No	NM	NM	MedQA: 85.4 on MedMCQA: 72.3, PubMedQA: 75.0	NM	MedQA, MedMCQA, PubMedQA, MMLU and human evaluation
	Yizhen Luo et al. 2023	Fine-tuned	No	NM	NM	USMLE: 50.4, MedMCQA: 51.4, PubMedQA: 76.1	NM	USMLE, MedMCQA, PubMedQA
	Hyunjae Kim 2024	Fine-tuned	No	8 × 80G A100 GPUs	36	MedQA: 70.6 USMLE: 70.3 Medbullets-4: 58.7 Medbullets-5: 52.9 MedMCQA: 60.6 MMLU: 70.5	NM	MedQA, USMLE, Medbullets-4, Medbullets-5, MedMCQA, MMLU
Semantic similarity	David Chang et al. 2021	Fine-tuned	No	NM	NM	0.868	NM	MedSTS- 2019 n2c2/OHNLP

NM: Not mentioned, GLUE: General Language Understanding Evaluation, SQuAD: Stanford Question Answering Dataset, DL: Deep learning, SWAG: Situation with Adversarial Generations, NER: named entity recognition, LSTM: bidirectional long short-term memory, SOTA: State-of-the-art their base model [18, 24], 1 study used *RoBERTa* [17], and 2 studies did not mention their base model [19, 37]. It is important to note that these studies utilized these proprietary model architectures with zero-weight therefore needed to be pre-trained.

Conversational tasks

In 8 out of the 13 studies, models were applied to conversational tasks, including medical V Q A(MedVQA) and Medical Report Generation (MRG) [21, 26, 27, 30, 31, 33–35]. The reported accuracies for these tasks ranged from 79% to 89.5%, with *Bio-MedGPT* model achieving the highest accuracy [27]. Only 1 out of 8 studies referenced *BERT* as its base model, while the others did not mention them [35].

Fine-tuned models

In a total of 11 studies, pre-existing GenAIs were modified and refined to better handle specific tasks such as categorizing data into different labels, learning to engage in effective conversations, and analyzing semantic similarity between texts [6, 16, 20, 22, 23, 25, 28, 29, 32, 36, 38].

Classification tasks

A total of 4 studies classified information, into structured and unstructured data, using fine-tuned models like *Clinical KB-BERT* [23, 25, 28, 29]. Among them, only 1 study reported the accuracy of its fine-tuned model, which was 99.53% [29]. All of the studies reported F-1 scores ranging from 0.54 to 0.99, with the highest score coming from the same study that reported the highest accuracy [29]. Among the 4 studies, 2 studies utilized *BERT* as their base model [28, 29], 1 study used pre-trained *Gatortron* [25], and another employed *GLoVe* [23].

Conversational tasks

Of the 11 studies, 6 focused on fine-tuning models for healthcare-related conversations and delivering information to patients [6, 16, 20, 32, 36, 38]. Accuracies were reported by 5 studies [6, 16, 20, 32, 36] ranging from 70.6% to 99.9%, with the highest accuracy achieved by a model named *OphGLM* designed for detecting ophthalmic diseases [16]. The study which did not report the accuracy provided the precision of 0.906 [38]. All 6 studies mentioned their base models used, such as *LLaVa*, *PaLM-2*, and *Mistral-7B-v0.1*.

Semantic similarity tasks

Only 1 study developed a fine-tuned model, *ClinicalBERT_all*, which combined linguistic and domain knowledge to assess textual similarity between clinical notes [22]. This model achieved a Pearson correlation of 0.882.

Multimodal models

A total of 8 out of 24 studies focused on multimodal models, which integrate diverse data types like images and text to improve model performance [16, 21, 25, 27, 30, 31, 33, 36]. All but one study reported accuracies achieved by their models on different datasets ranging from 80% to 99.9% [16, 21, 27, 30, 31, 33, 36]. The highest accuracy was acquired by *OphGLM* model tailored for ophthalmology disease diagnosis [16]. The study that did not report accuracy instead provided an F-1 score of 0.545 and a precision of 0.467 [25]. The overall accuracies reported in the studies is presented in Fig. 2.

GPUs

GPUs were employed to accelerate model training and enhance computational efficiency in 14 out of 24 studies [18, 19, 21, 24, 26–28, 30–32, 34–36, 38]. The reported training times varied between 12 and 552 h, depending on the complexity of the task. Only 1 of the 24 studies disclosed that training its model, *MEDITRON-70B*, resulted in a carbon emission of 486 $kgCO_2$, while the remaining 23 studies did not provide such data [34].

Benchmarks

All of the studies included in this scoping review reported the benchmarks they used to assess the performance of their models. A total of 17 out of the 24 studies evaluated their models using standard benchmarks [6, 13, 16, 17, 20–22, 26, 27, 30–32, 34–38], while the remaining 7 studies opted for custom benchmarks [18, 19, 23–25, 28, 29] tailored to their specific needs. The most commonly used standard benchmark datasets were VQA, RAD, and Slake, each used by 5 studies [21, 27, 30, 33, 36]. Additionally, another 4 studies utilized the MedMQA datasets [6, 32, 34]. The reporting completeness of all technical parameters are summarized in Table 4.

Comparative analysis

Across studies, healthcare applications were clustered into three main domains: (1) classification tasks (e.g., disease severity prediction, specialty identification), (2) conversational and question-answering tasks (e.g., medical VQA, USMLE style question answering), and (3) semantic similarity or information retrieval tasks. Conversational applications were most common (14/24 studies), reflecting growing interest in patient-facing and decision-support systems.

Fine-tuned models generally achieved higher task-specific performance with reduced training time compared to large-scale pre-trained models, suggesting greater efficiency for domain adaptation. Multimodal models, although fewer in number, demonstrated consistently strong performance in tasks integrating image and text

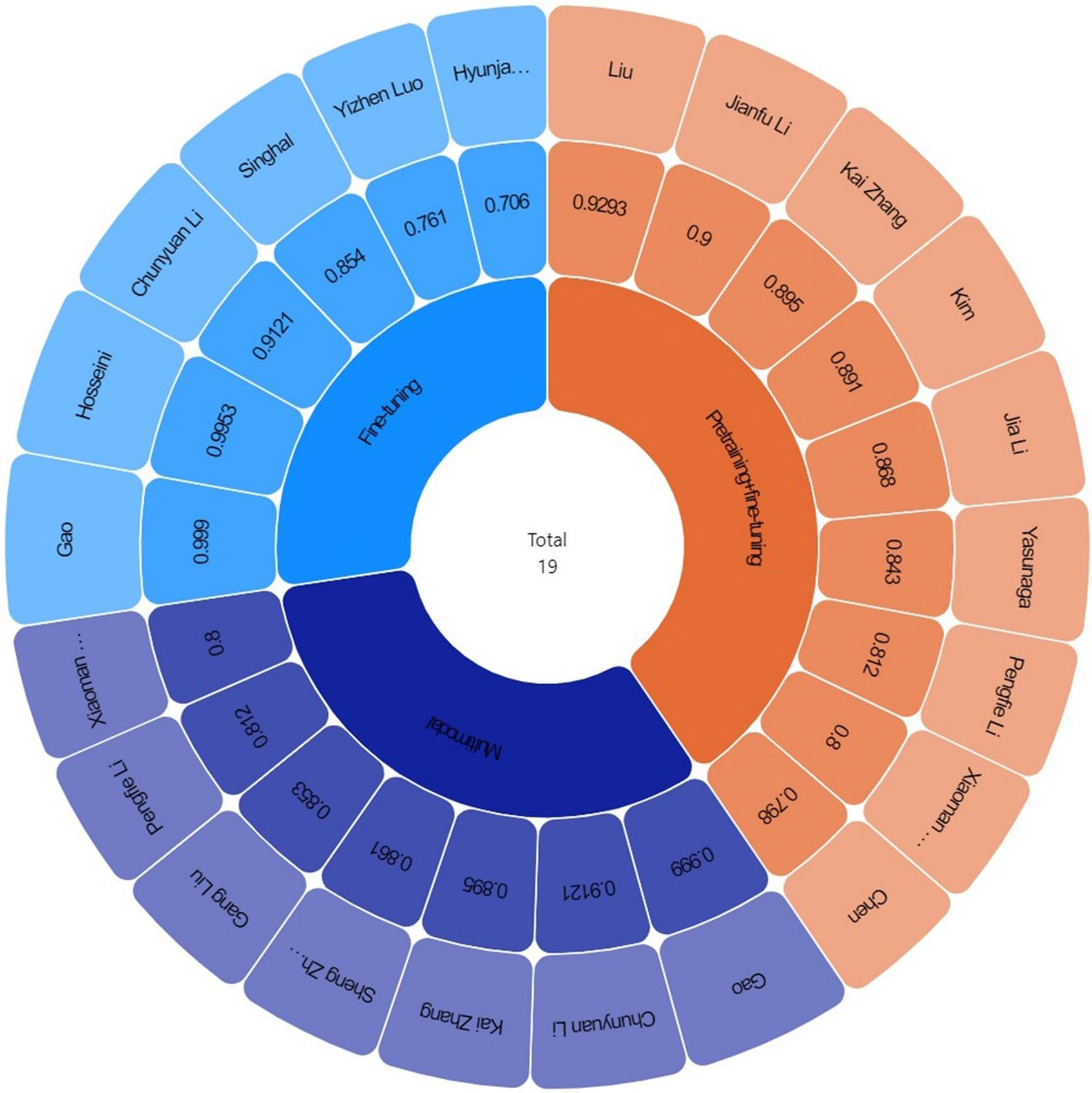

Fig. 2 Overall accuracies of models reported in each study. The wheel chart categorizes GenAI models in the rings based on color coding, with pre-trained plus fine-tuned (orange), fine-tuned (blue) and multimodal models (purple). The percentages on the outer ring represent the accuracies reported in the corresponding studies. The studies which did not report the accuracies are not included in the chart

Table 4 Reporting completeness of technical parameters among included studies

Characteristic	Completeness out of 24 studies
Sample size	21
Multimodal	8
GPU	14
Carbon emissions	1
Training hours	9
Benchmark	22

Graphical Processing Unit; GPU

data, highlighting their potential for complex clinical workflows.

Discussion

AI has been rapidly incorporated in different domains of healthcare, especially through the general purpose models [39]. Even though models such as *ChatGPT*, *Gemini*, *and Llama* are easily accessible yet they possess limited utility in healthcare [40]. Additionally, they often struggle to achieve high accuracies when applied to healthcare settings due to the unstructured nature of medical data [40]. As a result, domain-specific pre-trained and fine-tuned models have emerged to achieve greater accuracies across various benchmarks. This review highlights the significance of these domain specific models, especially in classification and conversational tasks, where most models consistently demonstrated strong performance, with accuracies ranging from 70.6% to 99.53% [29, 32].

Most of the studies included in this scoping review were published after 2019, following the introduction of GenAIs, marked by the release of Google's *BERT* model [41]. The characteristics of the included studies reveal a predominance of research conducted in China and the USA. This highlight, regional disparities in the use of AI for healthcare practices. Additionally, the use of diverse data sources for training and testing models raises concerns about data quality as variability among these sources may introduce potential biases and impact the generalizability of the findings.

The review also emphasizes that the fine-tuning process alone can lead to higher performance gains. A notable comparison involves the pre-trained *MEDITRON-70B* model, which required 332 h of training on 128 (A100,80GB) GPUs and achieved only 52% accuracy on the MedQA benchmark [34]. In contrast, the fine-tuned *Meerkat-7B* model attained 70.6% accuracy with just 36 h of training on 8 (A100 80GB) GPUs [32]. This suggests that a focus on fine-tuning cannot only enhance performance but also significantly reduce training time.

GenAI has been utilized for different healthcare problems for the benefit of simplifying it we stratified the tasks into either classification, conversational or semantic search (similarity between notes). The most common task for which GenAI were used in healthcare was conversational tasks, with 14 out of the 24 studies focusing on this area. Conversational tasks are crucial in healthcare because they involve direct interaction with patients or healthcare providers. They can be used for purposes like answering medical questions from patients or professionals generating summaries of patient records facilitating communication between healthcare provider. Fine-tuned models were particularly successful in conversational tasks, achieving accuracies ranging from 70.6% to 99.9% [16, 32].

The findings of the studies in this review also indicate that multimodal models were used in only a few studies, even though they have a lot of utility especially with unstructured data. Given the diverse nature of healthcare data, which often includes multiple types of information such as medical images, patient records, and clinical notes, multimodal models are particularly well-suited for handling such complexity [42]. The under-reporting of multimodal models could be attributed to the fact that they have just started to gain popularity amongst researchers [43].

Training GenAIs is a resource-intensive task requiring enormous computational resources that consumes a mammoth of energy [44]. Such consumption of energy may significantly impact the environment, leading to an increase in carbon footprint, negatively affecting Sustainable Developmental Goal (SDG) 13 [45]. One study in this review reported that training its model, *MEDITRON-70B*, resulted in a carbon emission of 486 $kgCO_2$ [34]. Since the majority of studies have not disclosed their carbon footprints, future researchers are encouraged to include this metric to better assess the environmental implications of GenAI training. Reporting this metric can also help future studies to compare the carbon emissions of unimodal and multimodal models. This analysis will help clarify the environmental impact of both approaches, enabling informed decisions on which models to adopt in future.

Another crucial element of this review was highlighting the benchmarks used to evaluate performance of different GenAIs. Benchmarks are any resource that has been published explicitly as a dataset and can be used for evaluation of AI models through analyzing specific metrics such as accuracy and precision [46]. Benchmark datasets like MedQA, VQA, and RAD provide a standardized method for comparison of different AI models and guide decisions for real-world applications. All the studies in our review assessed their models using either standardized or custom benchmarks hence emphasizing the importance of such evaluation benchmarks. MedQA emerged as the most frequently utilized benchmark dataset for general-purpose healthcare GenAIs. It is recommended that researchers adopt MedQA as a standardized benchmark dataset to minimize heterogeneity and enhance comparability amongst studies. It is important to note that there are currently no benchmark datasets available for dental-related tasks, revealing a gap in the field for evaluating dental-specific AI models.

The data used to train GenAIs is derived from the internet, which may lead to contamination and overlapping of data between training and testing sets [47]. The presence of data contamination can lead to overestimated performance metrics, as the model might inadvertently learn from test data that was leaked into the training set [47]. This compromises the integrity of the evaluation, making it difficult to gauge the model's true capabilities. To address these challenges, incorporating unseen data may reduce the impact of data contamination and lead to unbiased performance measures.

The integration of Gen AI into clinical workflows also brings significant ethical and privacy concerns that must be addressed to safeguard patient information [48]. Although the models in this review showed high performance these systems often rely on sensitive data from EHRs and VQA datasets. Using this information creates a risk of identification, where an individual's identity could be accidentally revealed by combining specific medical history with unique imaging data [49].

While the majority of the models described in this review remain at the stage of internal validation or proof-of-concept they must progress towards rigorous external validation, prospective clinical testing, and regulatory approval to ensure reliability and safety [50]. However, before large-scale deployments, the aforementioned challenges should be systematically addressed. Conducting small-scale pilot deployments can provide valuable insights into operational feasibility, ethical considerations, and real-world performance, allowing researchers and developers to refine these systems before broader clinical integration [51].

Regarding the strengths of this review, to the best of our knowledge, it is the first study to comprehensively report on all types of models within a single study. It also compares and analyzes the different training approaches employed by various GenAI models. Furthermore, it has taken a significant step towards advancing sustainable AI by highlighting the importance of reporting the carbon footprint associated with AI development.

Several important limitations should be considered when interpreting the findings of this review. First, regarding the search strategy, the study was confined to only three databases, with the notable exclusion of Web of Science due to access restrictions. This limited database coverage may have resulted in missing relevant studies, potentially affecting the comprehensiveness of our findings. Second, although the data synthesis form was systematically developed based on the authors' collective observations during the review process, it was not piloted prior to use. Furthermore, while the authors initially planned to conduct a meta-analysis of performance metrics, this proved unfeasible due to substantial methodological heterogeneity across the included studies. This heterogeneity prevented meaningful statistical comparison and synthesis of the quantitative results. The scope of our review was further constrained by specific exclusion criteria related to AI architectures. We omitted certain technical approaches, specifically Retrieval-Augmented Generation (RAG) and prompt engineering techniques, as these methods focus on model utilization rather than model training but may have an impact in GenAI utility.

Future reviews should explore a broader and diverse spectrum of AI architectures, including specialized foundation models, self-supervised learning frameworks, and reinforcement learning-based approaches to evaluate their potential in healthcare applications. They could also focus on studies exploring the integration of AI with wearable devices, real-time monitoring systems, or Internet of Medical Things (IoMT) platforms to understand practical clinical impact. In addition, reviews should address dental-specific benchmarks as they become available, which this study has identified as currently lacking in literature.

Conclusion

This scoping review highlights the critical role of AI in healthcare, particularly through pre-trained and fine-tuned models. It also notes the underutilization of multimodal models, which can better integrate diverse data types. However, the energy-intensive nature of training these models raises environmental concerns, emphasizing the need for carbon footprint reporting. The advancement of AI in healthcare is promising but requires careful consideration of model training techniques to ensure their effective and sustainable application.

Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
LLMs	Large Language Models
PRISMA-SCR	Preferred Reporting Items for Systematic Reviews and Meta–Analyses–Scoping Review
OSF	Open Science Framework
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
VQA	Visual Question Answering
EHR	Electronic Health Record
VO	Vaccine Ontology
UMLS	Unified Medical Language System
DRD	Diabetic Retinopathy Detection
GPU	Graphics Processing Unit
QA	Question Answering
MRG	Medical Report Generation
F-1	F–measure (Harmonic mean of precision and recall)
SDG	Sustainable Development Goal
MIMIC-III	Medical Information Mart for Intensive Care III
CT	Computed Tomography
NIHSS	National Institutes of Health Stroke Scale
CNER	Clinical Named Entity Recognition
NLP	Natural Language Processing
GCN	Graph Convolutional Network
CV	Computer Vision
CSCR	Chinese Stroke Clinical Records
GM	Generative Model
Med-VQA	Medical Visual Question Answering
MRG	Medical Report Generation
MedVInT	Medical Visual Instruction Tuning
DRD	Diabetic Retinopathy Detection
BLURB	Biomedical Language Understanding and Reasoning Benchmark
MMLU	Massive Multitask Language Understanding
MedQA	Medical Question Answering dataset
PubMedQA	PubMed Question Answering dataset
MedMCQA	Medical Multiple Choice Question Answering dataset
USMLE	United States Medical Licensing Examination
STS	Semantic Textual Similarity
GLUE	General Language Understanding Evaluation
SQuAD	Stanford Question Answering Dataset
SWAG	Situations With Adversarial Generations
NER	Named Entity Recognition
LSTM	Long Short–Term Memory
SOTA	State–of–the–Art

Supplementary Information

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Supplementary Material 1

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